

Enhancing IOT BotNet Detection through Hybrid Ensemble Learning

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Introduction to BotNets

- **Definition:** A botnet (short for "robot network") is a logical collection or group of Internet-connected devices (e.g., computers, smartphones, IoT devices) whose security has been breached.
- **Components:** Each compromised device is known as a "bot" or "zombie". These devices are controlled remotely by a cybercriminal, known as the "bot herder" or "botmaster," via Command and Control (C&C) software or servers.
- **Purpose (Malicious):** Botnets are used to carry out coordinated, large-scale malicious activities, including Distributed Denial-of-Service (DDoS) attacks, sending spam, stealing data, conducting financial fraud, and distributing malware (such as ransomware).

The Growing Threat Landscape

- **Botnet Attacks:** Coordinated networks of compromised devices executing malicious activities
- **Scale:** Cisco predicted attacks to double from 7.9M (2018) to 15.4M (2023)
- **Impact:** 60% of attacks target financial and telecom sectors

IoT Vulnerability Factors

- **Resource Constraints:** Limited CPU, memory, and processing power
- **Massive Scale:** Billions of connected devices creating attack surfaces
- **Heterogeneity:** Diverse device types with varying security implementations

Literature survey:

Paper Name	Algorithms Used	Description	Use in My Project
N-BaloT—Network-Based Detection of IoT Botnet Attacks Using Deep Autoencoders	Deep Autoencoders	Introduced the N-BaloT dataset and used autoencoders to detect anomalies in IoT network traffic	Provides the foundational dataset and anomaly detection approach; informs dataset selection and baseline method
CNN-based Detection (Karaca & Çetin, 2021)	CNN	Applied CNN for IoT traffic pattern recognition to detect botnet activities by spatial feature extraction	Spatial feature extraction techniques help improve feature engineering in ensemble approach
Efficient Malware Classification with Spiking Neural Networks: A Case Study on N-BaloT Dataset	Spiking Neural Networks (SNNs)	Bio-inspired, energy-efficient malware classification by modeling temporal dynamics of malware activity	Influences the design principles for energy-efficient and lightweight models

Paper Name	Algorithms Used	Description	Use in My Project
Anomaly detection in Internet of Things using feature selection and classification based on Logistic Regression and Artificial Neural Network on N-BaloT dataset	ANN with Feature Selection	Combines feature selection with classification models to enhance anomaly detection in IoT networks	Drives use of PCA and scaling techniques to improve feature selection and reduce dimensionality
Machine Learning-Based IoT-Botnet Attack Detection with Sequential Architecture	Sequential Machine Learning Models	Temporal relationship modeling for detecting botnet behavior based on sequential patterns	Directly informs our concept drift detection and adaptive learning approach
SkipGateNet: A Lightweight CNN-LSTM HybridModel With Learnable Skip Connections forEfficient Botnet Attack Detection in IoT	CNN-LSTM	Lightweight hybrid model achieving high accuracy (99.7%) in botnet detection	Sets performance benchmarks and motivates combining models for balanced efficiency and accuracy
A Horizontal Federated Learning Approach to IoT Malware Traffic Detection: An Empirical Evaluation with N-BaloT Dataset	Horizontal Federated Learning	Privacy-preserving distributed learning applied to IoT detection	Suggests future integration into our adaptive ensemble to enhance data privacy and scalability

Existing System: IoT BotScan Framework

Architecture Overview

- Data Preprocessing → Feature Engineering → Random Forest → Classification

Data Preprocessing

- - Duplicate and NaN removal
- - 80-20 train-test split
- - Standard and MinMax scaling

Feature Engineering

- - PCA with 95% variance retention

Model Performance

- - 99.94% KNN Accuracy
- - 99.55% Random Forest
- - 86.20% Best DL (CNN)
- - 5.5s RF Test Time

Existing System: Performance Analysis

Detailed Results Comparison

Machine Learning Models

- KNN: 99.94% Acc, High Test Time
- Random Forest: 99.55% Acc, 5.5s Test Time
- Extra Trees: 99.55% Acc, Low Test Time
- Decision Tree: 99.54% Acc, Low Test Time

Deep Learning Models

- CNN: 86.20% Acc, 20-25s Test Time
- LSTM: 85.30% Acc, 20-25s Test Time
- RNN: 82.10% Acc, 20-25s Test Time

Key Finding: Random Forest provides optimal balance of accuracy (99.55%) and efficiency (5.5s), making it ideal for IoT deployment

Existing System: Critical Limitations

Real-World Deployment Challenges

- 1. Adversarial Vulnerability: No defense against crafted evasion attacks
- 2. Static Architecture: Cannot adapt to evolving attack patterns
- 3. Limited Validation: Tested only on N-BaIoT dataset
- 4. Deployment Gaps: No real-time adaptation, limited scalability validation

Critical Gap: While IoT BotScan achieved 99.55% accuracy in controlled conditions, it lacks the robustness and adaptability required for real-world IoT security.

Research Gaps

Primary Research Goals

- 1. Adversarial Robustness: Adversarial training for ensemble models
- 2. Adaptive Learning: Automated concept drift detection & incremental learning
- 3. Comprehensive Validation: Multi-dataset evaluation & cross-environment testing
- 4. Production Deployment: Real-time processing optimization & scalability validation

Innovation Focus: Transform static laboratory model into dynamic, production-ready defense system capable of handling real-world IoT security challenges

Proposed System: Resilient Adaptive Architecture

Core Innovation: Stacking Ensemble

- - Random Forest + XGBoost + LightGBM → Meta-Learner → Robust Prediction

1. Adversarial Training

- - FGSM, PGD, C&W attack generation
- - Robust training with adversarial examples
- - Gradient masking defense mechanisms

2. Concept Drift Detection

- - Kolmogorov-Smirnov tests(KS Test), Page-Hinkley change detection
- - Performance degradation monitoring

3. Multi-Dataset Integration

- **N-BaIoT (Baseline) – 8.14gb** -This dataset addresses the lack of public botnet datasets, especially for the IoT. It suggests real traffic data, gathered from 9 commercial IoT devices authentically infected by Mirai and BASHLITE.
- **IoT-23 (Validation) – 8.7gb** - It has 20 malware captures executed in IoT devices, and 3 captures for benign IoT devices traffic
- **BoT-IoT (Diverse attacks) – 15gb** – BoT-IoT dataset was created by designing a realistic network environment in the Cyber Range Lab

Technical Implementation Details

Stacking Ensemble Design

- Random Forest: Proven performance, low overfitting
- XGBoost: Gradient boosting, feature importance
- LightGBM: Fast training, memory efficiency

Adversarial Training Protocol

- Phase 1: Clean data training
- Phase 2: Generate adversarial examples (FGSM, PGD, C&W)
- Phase 3: Joint training (70:30 clean:adv)
- Phase 4: Robust meta-learner training

Concept Drift Adaptation

- - Monitoring: Sliding window statistical analysis
- - Detection: Configurable thresholds
- - Response: Incremental retraining with new data